MultiRate.jl: findings

Partially implicit MIS methods for the compressible Euler equations with gravity on DGSEM, terrain-following FV and cut-cell FV

November 2026

Abstract

This report collects the results of the numerical experiments with MultiRate.jl in the order they were obtained: background-free discretizations, the Agnesi hills, gravity waves, the MIS tableaus, linear acoustics with advection after the MISDG experiments, the linearized production models and a matrix-free analysis of the linearized step, and Lawson RK with the MIS fast integrators. The recurring result is that the slow part $S = N - L$ of a split-explicit step must not contain terms whose Courant number with the large step is large. Two instances are identified: the vertical transport of horizontal momentum by the fast vertical velocity (Agnesi a100 and the long-channel gravity wave, $U\Delta t/\Delta z = 13$–22), and a grid-scale mode on the steep flanks of the a1 hill (preliminary). The markdown version of these findings is `docs/FINDINGS.md`; the MISDG talks are in `docs/talks/`.

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1 The method

The semi-discrete system is $q' = N(q)$. At the start of a large step the fast linear operator L is frozen at q^n (in the runs it is re-frozen every 10 steps), and $S(q) = N(q) - Lq$ is the slow part. Stage i of an MIS method solves the fast system

$$Z' = LZ + \sum_{j<i} \frac{\gamma_{ij}}{d_i \Delta t}\, \Delta Y_j + \sum_{j \le i} \frac{\beta_{ij}}{d_i}\, S(Y_j), \qquad Z(0) = q^n + \sum_{j<i} \alpha_{ij} \Delta Y_j, \tag{1}$$

over $[0, d_i\Delta t]$, $d_i = \sum_j \beta_{ij}$, with $n_i = \max(1, \mathrm{round}(d_i\, n_{\mathrm{sub}}))$ substeps, and sets $Y_{i+1} = Z(d_i\Delta t)$, $\Delta Y_i = Y_{i+1} - q^n$. The paper form uses $(s+1)\times(s+1)$ matrices; MultiRate stores $\beta = P[2{:}\mathrm{end}, 1{:}\mathrm{end}-1]$ and $\alpha, \gamma = P[2{:}\mathrm{end}, 2{:}\mathrm{end}]$ (`mis_from_paper`). For $L = 0$ the method is the Runge–Kutta method with $B = (I - \alpha - \gamma)^{-1}\beta$.

The fast integrators are `:verlet` (Störmer–Verlet, thermodynamic kick and momentum), `:fb` (forward–backward), `:hevi` (horizontally explicit, vertically implicit), for FV the Strang composition of horizontal Verlet half steps around a vertical Crank–Nicolson step, `:hevi_verlet` (DGSEM), `:wrf` (WRF's off-centred vertical step with divergence damping) and, for cut cells, a Rosenbrock-W AMF step (`:rosenbrock_amf`).

The spatial discretizations are DGSEM (degree 4, LMARS/entropy-conservative volume fluxes), a terrain-following Arakawa C-grid FV with a Gal-Chen vertical coordinate, and a cut-cell C-grid FV. All production runs use MIS4 with $n_{\mathrm{sub}} = 12$.

2 Discretizations without a background state

- DGSEM uses the pointwise fast operator L (frozen $\bar\theta$ and $\gamma p/\Theta$, as in the HEVI Jacobian of Artiano et al.); terrain-following FV the conservative `:flux` form of the Θ equation; cut cells `:donor`. None of them uses $\bar\theta(z)$.
- FV without a hydrostatic reference $\bar p, \bar\rho$ (`FV_HYDRO_REFERENCE[] = false`, the default): the momentum equations use full p and ρ. The analytic hydrostatic initial state, sampled where the operator places the values (cut cells: the full-cell centre), is balanced to $O(\Delta z^2)$ (test: residual ratio > 1.9 between $n_z = 6$ and 12).
- At rest, 6 h on the a10 grid, $\max|w|$: terrain FV $4.1\cdot10^{-3}$ m/s (10^{-11} with the reference), cut cells $9\cdot10^{-3}$ m/s ($3\cdot10^{-5}$ with the reference).
- **Ground pressure in the Gal-Chen correction** $-\partial_\eta(z_x p)$: the ground-face p must be extrapolated from the face *means* (`fv_p_sigma`, $\frac{3}{2}p_1 - p_3 + \frac{1}{2}p_4$), so that its $O(\Delta\eta^2)$ error matches that of the interior means and cancels. Taking the lowest cell centre is off by $\rho g\Delta z/2$, an $O(1)$ error ($|U_t| \approx 0.14$ at rest, independent of resolution); pointwise 2nd/3rd-order extrapolation is only first order. The fast operator includes the ground face the same way.
- The flux form has a weak growing mode at rest on terrain (the x-flux carries θ along the sloped σ-surfaces, while the σ-correction uses 4-point averaged U; $\approx 5\cdot10^{-6}\,\mathrm{s}^{-1}$ on a coarse test hill, growing with horizontal resolution). It does not affect the Agnesi runs. `:advective` (state θ, adjoint pressure gradient) removes it at the price of Θ conservation.

3 Agnesi hills

Gallus & Klemp (2000): $h = 400$ m, $U = 10$ m/s, $N = 0.01\,\mathrm{s}^{-1}$, half-widths $a = 1, 10, 100$ km, periodic channel of width $40a$, $T = 21.6\,a/U$; FV 200×64 cells, DGSEM 50×16 elements of degree 4. $\max|w|$ (m/s) and the θ' range (K) at the final time:

	DGSEM `hevi_verlet`	FV terrain-following `hevi`	FV cut cells `rosenbrock_amf`
large steps	694	400	565
a1	2.77, $-1.22\ldots0.93$	2.54, $-1.09\ldots0.92$ ($2\Delta x$ noise)	2.14, $-1.08\ldots0.91$
a10	0.69, $-1.48\ldots1.65$	0.56, $-1.39\ldots1.55$	0.58, $-1.43\ldots1.58$
a100	0.064, $-1.49\ldots1.70$	0.056, $-1.40\ldots1.57$	0.057, $-1.43\ldots1.61$

- **a100 and the vertical momentum transport.** The vertical transport of horizontal momentum by the fast vertical mass flux, $-\partial_z(\bar u\, W')$, must be in the implicit vertical L (`DG_VADV_FAST`, `FV_VADV_FAST`). Left in S it couples the large step to the vertical acoustics

through the vertical Courant number $U\Delta t/\Delta z$ (a1 0.1, a10 1.3, a100 13 for DGSEM, 22 for FV), and a100 blows up. $\partial N/\partial q$ has no growing mode: this is a stability limit of the explicit part of MIS, not of the spatial discretization. The term is put into L for $U\Delta t/\Delta z > 4$. A split-form linearization of L (`DG_SPLIT_FAST`) makes $N - L$ vanish at rest but changes neither stability nor solution; the simple pointwise L is kept. For FV, L must in addition be skew.

- **DGSEM with `hevi_verlet`** runs all three hills at the regular horizontal acoustic step (a100: $d\tau = 26\,\mathrm{s}$, $\Delta t = 313\,\mathrm{s}$), about 16 times fewer steps than `hevi`.
- **Cut cells:** `hevi` and `rosenbrock_amf` give the same solutions; AMF also runs at the large steps of the MATLAB reference ($\Delta t = 20\,\mathrm{s}$ for a1, $100\,\mathrm{s}$ for a10). Its fast-step CFL is 0.65 of the regular-grid limit (explicit horizontal acoustics inside the W-method). Open: a warm layer of $\approx 0.35\,\mathrm{K}$ in the lowest cells downstream of the hill (a10).
- **a1, terrain FV:** $2\Delta x$ noise near the ground at the default step $\Delta t = 5.4\,\mathrm{s}$. It disappears with half the slow step (`NSUB=6`, same $d\tau$, $\max|w| = 2.41$), so it is a slow-step limit for terrain terms in $N - L$, not a spatial defect. It is also present with the hydrostatic reference and with the `:advective` form. At twice the resolution the default step blows up; `NSUB=6` is stable (2.68). See Section 8.
- Runs with $U \neq 0$ without the reference drift in mass by $\approx 2.5 \cdot 10^{-5}$ ($3.7 \cdot 10^{-7}$ with it), attributed to the sponge relaxing to a q_0 that is not a discrete equilibrium; not investigated further.

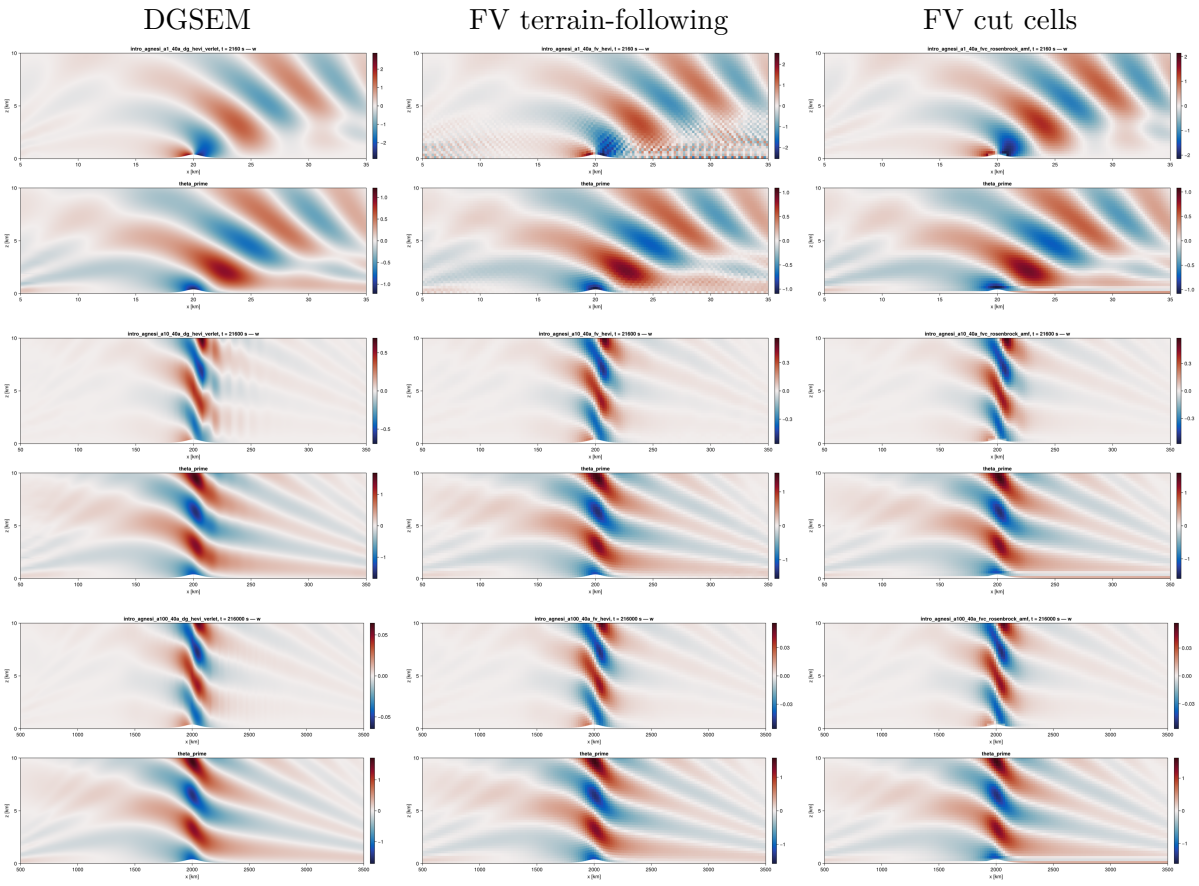


Figure 1: Agnesi hills a1 (top), a10, a100 (bottom): w and θ' at the final time.

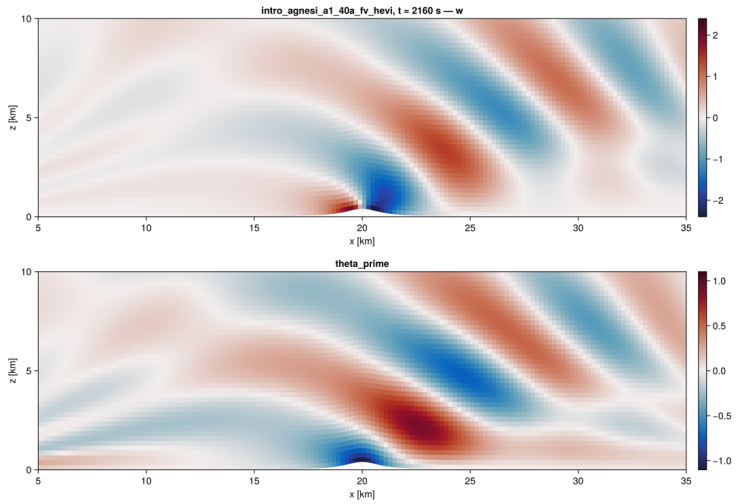


Figure 2: Terrain-following FV, a1, half the slow step (`NSUB=6`): the ground noise of the default step is gone.

4 Gravity waves

Skamarock & Klemp (1994), flat ground:
`euler2d_gravity_wave.jl` (300 km) and `euler2d_gravity_wave_long.jl` (6000 km, $a = 100$ km, $\Delta x = 20$ km, $\Delta z = 625$ m, $U = 20$ m/s, $T = 60000$ s), MIS4, `hevi`, $n_{\text{sub}} = 12$.

- 300 km: DGSEM and FV symmetric (θ' maxima $2.73/2.72{\cdot}10^{-3}$ K DGSEM, $3.78/3.32{\cdot}10^{-3}$ K FV).
- 6000 km, FV at the default step $\Delta t = 540$ s ($N\Delta t \approx 5.4$, $U\Delta t/\Delta x = 0.54$, $c\Delta t/\Delta x \approx 9.5$): asymmetric packets ($\theta'$ maxima 2.2 and $9.0{\cdot}10^{-3}$ K) with grid-scale noise between them. The same with the hydrostatic reference, with the commit before the background was removed and with both fast solvers. With `NSUB=3` ($\Delta t = 135$ s, same $d\tau$) the packets are symmetric, so the error comes from the slow step. DGSEM is clean.
- **Cause and fix** (Section 7): $-\partial_z(\bar{u}W')$ was in S although $U\Delta t/\Delta z \approx 17$; the production drivers did not apply the vadv criterion of the Agnesi driver. With `set_vadv!` (now in all drivers) the default step gives θ' maxima $5.34/5.29{\cdot}10^{-3}$ K, asymmetry $4.4{\cdot}10^{-4}$ K ($6.9{\cdot}10^{-3}$ K before). The 300 km case ($U\Delta t/\Delta z = 2.1$) and DGSEM (0.6) are unchanged.
- The uniform $\approx 10^{-3}$ K warm offset of θ' is also present for a 100 times smaller perturbation and for $U = 0$: a diagnostic offset of θ' against $\bar{\theta}(z)$ at the cell centres, not part of the wave.

5 MIS tableaus

`TimeStepping.mis_observed_order` computes the order with exact fast stages (augmented exponential) on a problem where L and S do not commute, $L = \begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}$, $G(q) = (0.5\sin q_1 + q_2^2,\ -0.3q_1q_2 + \cos q_1)$; `rk_order` gives the order for $L = 0$.

	WSRK3	MIS2	MIS3C	MIS4	MIS4a	MIS_KW3	MIS54
RK order ($L = 0$)	2	2	3	3	3	3	4
MIS order	2	2	**2**	3	3	3	4
fast time $\sum_i d_i$	1.83	1.19	1.19	1.44	1.47	1.00	1.88

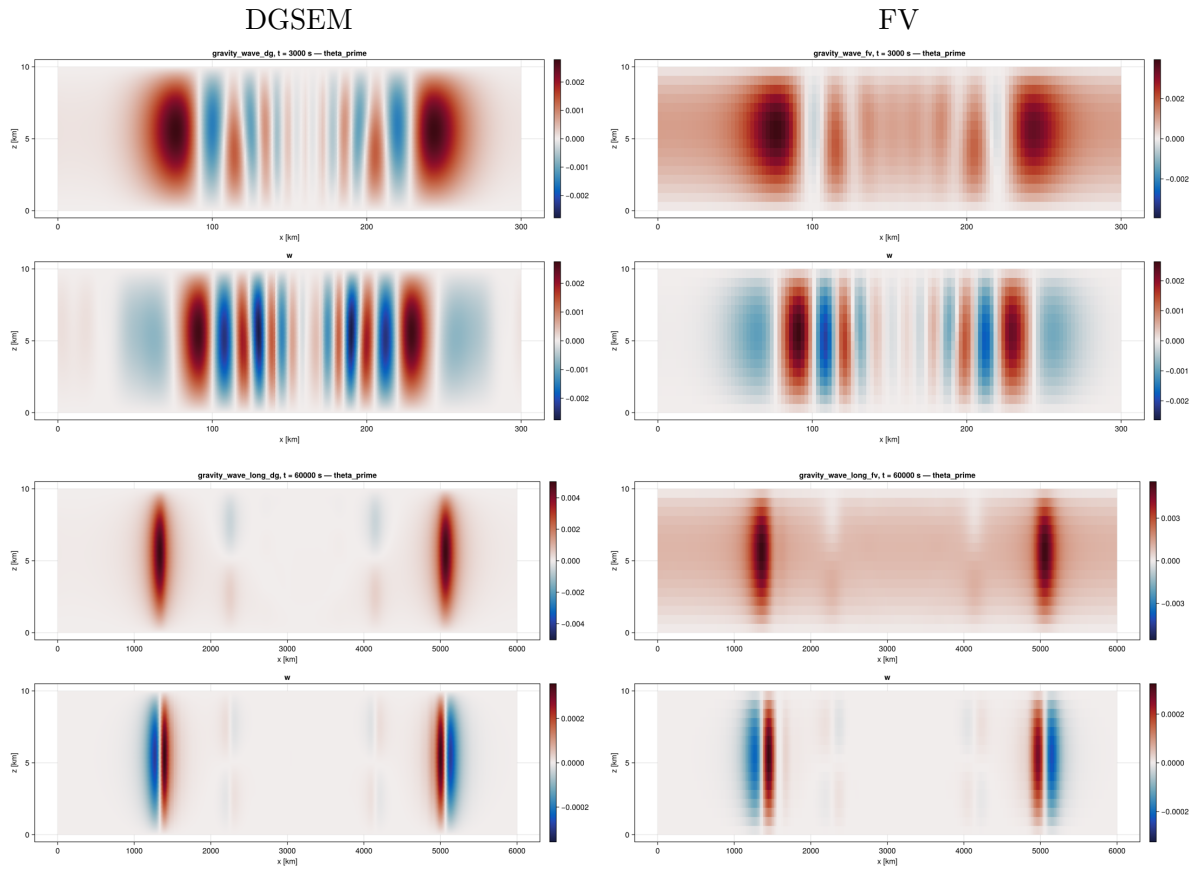


Figure 3: Gravity waves, θ' and w: 300 km channel at $t = 3000$ s (top), 6000 km channel at $t = 60000$ s (bottom; FV with the vertical momentum transport in L).

- MIS3C (coefficients as in ClimaTimeSteppers.jl) is only second order once $L \neq 0$: a coupling condition fails. To be checked against Knoth & Wensch (2014).
- MIS_KW3 is built from the Knoth–Wolke RK3 tableau; MIS54 is from Bauer & Knoth (2021), with the coefficients of MISDG. MIS54 is fourth order and runs in the FV gravity-wave test.

6 Linear acoustics with advection

`MultiRate.LinearAnalysis` treats a linear problem $q' = Nq$ with fast part L as a model of the TimeStepping interface. `step_matrix` runs the production `mis_step!` and fast loops on unit vectors, or uses exact fast stages. A Fourier mode with complex symbol A becomes the real model $\begin{pmatrix} \Re A & -\Im A \\ \Im A & \Re A \end{pmatrix}$. Script: `examples/linear_analysis/mis_stability.jl`. These are the experiments of the MISDG talk (`docs/talks/misdg`).

- **Knoth–Wensch test** (C-grid, 2nd-order acoustics, 3rd-order upwind advection, exact fast stages): pure advection is stable to $C_{\text{adv}} = 1.63$ (3-stage methods) and 2.06 (MIS54). Undamped WSRK3 is unstable from $C_{\text{sound}} \approx 3.2$, MIS_KW3 from ≈ 2 (and for any C_{sound} at $C_{\text{adv}} = 1$). MIS4a and MIS54 are stable on the whole box $C_{\text{sound}} \leq 12$.
- **MISDG grid** (4th-order staggered acoustics, 5th-order upwind advection, $C_{\text{adv}} = C_{\text{sound}}/7$), exact fast stages: WSRK3 stable to $C_{\text{sound}} = 2.75$, MIS4 beyond 14, MIS4a to 12, MIS54 to 14. MIS_KW3 is weakly unstable for every C_{sound} ($\rho - 1 = 7 \cdot 10^{-6}$ at $C_{\text{sound}} = 1$), as in MISDG.

- **Fast loops**, $n_{\text{sub}} = 12$: `:verlet` and `:fb` converge to the exact-stage result as $d\tau \to 0$ (orders 2 and 1). At $C_{\text{sound}} = 10$ the Verlet substep exceeds its own limit for every tableau. `:fb` makes MIS4 and MIS54 weakly unstable at $C_{\text{sound}} = 3$–7, `:verlet` does not. WRF divergence damping (`:wrf`) stabilizes WSRK3 up to $C_{\text{sound}} = 7$.
- **Pulse convergence** (1D pressure pulse, exact fast stages): orders 2 (WSRK3), 3 (MIS4a), 4 (MIS54).
- Consequence: acoustics with advection does not explain the slow-step limits of the Agnesi and gravity-wave runs; MIS4 and MIS4a are stable far beyond the Courant numbers of those runs.

MISDG grid, $C_{\text{adv}} = C_{\text{sound}}/7$	WSRK3	MIS4	MIS4a	MIS54
stable up to C_{sound} (exact fast stages)	2.75	> 14	12	14
with `:wrf`, $n_{\text{sub}} = 12$	7	7	7	7
pulse convergence order	2	3	3	4

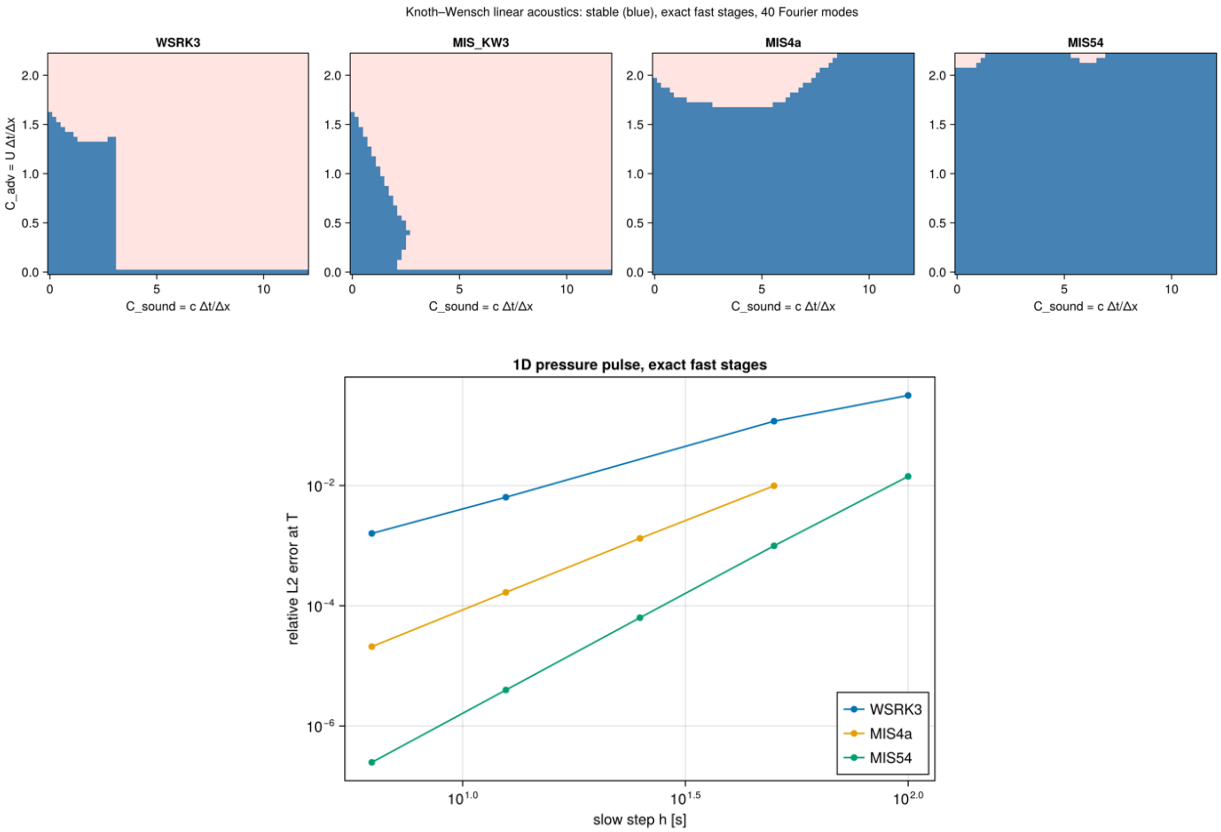


Figure 4: Top: Knoth–Wensch stability regions over $(C_{\text{sound}}, C_{\text{adv}})$, stable in blue, exact fast stages. Bottom: 1D pressure pulse, relative L^2 error against the slow step.

7 Linearized production models: the long-channel failure

Script `examples/linear_analysis/fv_gravity_wave_split.jl`. The flat FV model is invariant under shifts in x. `LinearAnalysis.linearized_mode_model` builds, per horizontal Fourier mode, the 63×63 column blocks of the Jacobian J of `rhs_nonlinear!` (central differences), of the frozen fast operator L and of its vertical part; `step_matrix` runs the production MIS step on them. The eigenvalues of all blocks together equal those of the full fast matrix (test).

Steps that led to the cause.

1. The real FV run is linear in the amplitude: with $\Delta\theta = 10^{-4}$ instead of 10^{-2} the asymmetry scales by 10^{-2}. With $U = 0$ it vanishes (10^{-5} K).
2. A constant-coefficient model on the FV grids (`boussinesq_symbols`: compressible Boussinesq, C-grid, Lorenz grid, rigid lid, buoyancy and acoustics in L, advection in S) is accurate at $\Delta t = 540$ s (error $1.5 \cdot 10^{-4}$ of a $4.5 \cdot 10^{-3}$ K packet). Moving the advection of any rows between L and S changes this by at most 1.4 %. The failure is not in the generic acoustic–gravity–advection problem.
3. The linearized production FV reproduces the nonlinear run (θ' maxima $1.319/8.017 \cdot 10^{-3}$ K against $1.320/8.022 \cdot 10^{-3}$ K). With exact fast stages the error remains, so it is the slow-step coupling; the `:hevi` loop adds instability ($\max|\lambda| = 1.15$ at $4\Delta x$–$8\Delta x$).
4. Moving rows of S into L (exact fast stages): the horizontal-momentum rows reduce the error from $1.67 \cdot 10^{-3}$ to $2.8 \cdot 10^{-5}$ K, the Θ rows do nothing; writing L's mass and Θ fluxes with the perturbation momentum does nothing either.
5. The stiff part of the momentum rows of S is $-\partial_z(\bar{u}W')$ with coefficient $\bar{u}\Delta t/\Delta z \approx 17$, the term `FV_VADV_FAST` puts into L.

L	fast	Δt [s]	packets [10^{-3} K]	max err [K]	$\max\|\lambda\|$
as in the driver (no vadv)	exact	540	6.13 / 2.80	$1.7 \cdot 10^{-3}$	1.0000
	`:hevi`	540	1.36 / 7.99	$4.0 \cdot 10^{-3}$	1.151
	`:hevi`	135	4.94 / 3.91	$5.6 \cdot 10^{-4}$	1.0034
vertical u-transport in L	exact	540	4.55 / 4.38	$1.2 \cdot 10^{-4}$	1.0000
	`:hevi`	540	4.42 / 4.38	$3.7 \cdot 10^{-4}$	1.0000
	`:hevi`	135	4.46 / 4.47	$2.0 \cdot 10^{-4}$	1.0000
all S momentum rows in L	`:hevi`	540	4.42 / 4.76	$4.5 \cdot 10^{-4}$	1.0015
exact linear solution			4.50 / 4.46		

MIS4, $n_{\mathrm{sub}} = 12$ (540 s) and 3 (135 s), modes $j \le 60$ of 300; $2.8 \cdot 10^{-5}$ K is the truncation floor. Regression test: a 30-column grid with the same $\Delta x, \Delta z, \Delta t$ has $\max|\lambda| > 1.1$ at $7.5\Delta x$ without vadv and < 1 with it.

Consequences.

- The vadv criterion ($U\Delta t/\Delta z > 4$) is a property of the step, not of a test case; every driver applies it through `set_vadv!`.
- Constant-coefficient model problems are good for the tableaus and fast loops but missed this; the linearized production operator is the reliable diagnostic.
- With `:wrf` instead of `:hevi` the default step is stable without vadv ($\max|\lambda| = 1.0014$) but still inaccurate ($3.6/5.7 \cdot 10^{-3}$ K).

8 Matrix-free analysis of the linearized step (preliminary)

`LinearAnalysis.linearized_step(model, qb, dt, tab; nsub, fast)`
applies the linearized production MIS step, $v \mapsto (\Phi(q_b + hv) - \Phi(q_b - hv))/2h$ with the fast operator frozen at q_b, to a vector. It needs no shift invariance. `arnoldi_eigs` (restarted Arnoldi in a weighted inner product) finds eigenvalues of largest modulus of such a map.

- On the a1 terrain-FV grid (51 200 unknowns) Arnoldi on the step itself does not converge: almost all eigenvalues lie close to the unit circle. Power iteration (repeated application) works.

- a1 FV linearized about the initial state, MIS4, `:hevi`: growth **1.027 per step** at the default step (`NSUB=12`, $\Delta t = 5.4\,\mathrm{s}$) and 1.0001 (neutral, a mode spread over the domain) at `NSUB=6`.
- The growing mode is a $2\Delta x$–$3\Delta x$ oscillation of U and W in the lowest three levels on the steep flanks (slope ± 0.25) near the crest, where the noise appears in the run. A first look at $(J - L)v$ on this mode shows the largest slow tendencies at the crest in the lowest two levels in the Θ, U and W rows. A plausible estimate of a fast terrain term left in S is $c\Delta t/\Delta z \cdot |z_x| \approx 7.7 \cdot 0.25 \approx 2$.
- Open: fast loop or slow-step coupling (same Δt with many substeps), and which terrain term of S is responsible. Terrain tests are paused for now.

9 Lawson RK with the MIS fast integrators as propagator

`TimeStepping.LawsonTableau` (`:LRK4` classical RK4, `:LRK3` Kutta, `:LRK2` midpoint) is selected by name like the MIS tableaus (environment variable `TABLE` in the drivers). With the fast flow φ_τ of the frozen L (including its constant forcing) and $S = N - L$,

$$Y_i = \varphi_{c_i\Delta t}(q^n) + \Delta t \sum_j a_{ij}\, e^{(c_i - c_j)\Delta t L} S(Y_j), \qquad q^{n+1} = \varphi_{\Delta t}(q^n) + \Delta t \sum_j b_j\, e^{(1-c_j)\Delta t L} S(Y_j). \quad (2)$$

The sums are evaluated by propagating through the distinct nodes in increasing order; $\varphi_{c\Delta t}(q^n)$ is computed once per node. Since φ_τ is affine, $\varphi_\tau(x + d) = \varphi_\tau(x) + e^{\tau L} d$, so the affine fast flow can be used throughout. The propagator is the fast loop of the MIS step (`fast_solve!` without slow forcing, $d\tau \approx \Delta t/n_{\mathrm{sub}}$). `mis_step!` dispatches on the tableau type, so `run_mis`, the drivers and `LinearAnalysis` work unchanged. Fast integration time per step: LRK4 $2.5\Delta t$, MIS4 $1.44\Delta t$.

- **Order** with exact propagators (non-commuting test): LRK2 2, LRK3 3, LRK4 4. With the Verlet loop the step converges to the exact one at order 2 in $d\tau$.
- **Linear stability** is limited by the advection Courant number, as for RK4: $C_{\mathrm{adv}} \le 1.75$ (LRK4), 1.62 (LRK3) on the Knoth–Wensch test (MIS54: 2.06); stable fraction of the box 0.78 (MIS54: 0.99). On the MISDG grid ($C_{\mathrm{adv}} = C_{\mathrm{sound}}/7$) LRK4 with Verlet and $n_{\mathrm{sub}} = 12$ is stable up to $C_{\mathrm{sound}} = 10$, where MIS4 and MIS54 with Verlet are already unstable; $|\rho| - 1 \approx 10^{-8}$ against 10^{-5} for MIS4.
- **Accuracy with exact propagators:** the pulse error is 190 times smaller than with MIS54 at the same step ($5.3 \cdot 10^{-6}$ against $9.9 \cdot 10^{-4}$), order 4. Lawson is exact for $S = 0$; MIS with $\alpha, \gamma \neq 0$ is not. For an unbalanced initial state (the gravity-wave θ' at constant pressure) all MIS tableaus damp or dephase the fast acoustic part by a Δt-independent amount ($4 \cdot 10^{-3}$ relative in the linear Boussinesq wave, also for the $k = 0$ mode); Lawson keeps it ($5 \cdot 10^{-13}$).
- **Accuracy with the real fast loops:** the propagator error enters directly. Pulse with Verlet: the LRK4 error is the second-order Verlet error (rates 2.00), the same as for MIS54. Nonlinear FV gravity wave (300 km, `:hevi`) against explicit RK4 on N:

$\max \lvert\theta' - \theta'_{\mathrm{RK4}}\rvert$ at $z = 5\,\mathrm{km}$ [K]	$\Delta t = 67\,\mathrm{s}$	34 s	17 s
MIS4	$1.1 \cdot 10^{-3}$	$8.6 \cdot 10^{-4}$	$6.0 \cdot 10^{-4}$
MIS54	$9.4 \cdot 10^{-4}$	$7.9 \cdot 10^{-4}$	$5.7 \cdot 10^{-4}$
LRK4	$1.6 \cdot 10^{-3}$	$1.7 \cdot 10^{-3}$	$3.7 \cdot 10^{-4}$

($\max \theta'_{\mathrm{RK4}} = 4.4 \cdot 10^{-3}\,\mathrm{K}$, $n_{\mathrm{sub}} = 12$.) All methods are dominated by the fast-loop error (vertical Crank–Nicolson at large vertical Courant numbers).

Consequence. With second-order fast loops Lawson brings no gain over MIS4; its fourth-order accuracy needs an accurate propagator, i.e. the high-order (Yoshida) fast integrator or an exact or high-order vertical solve. A MIS run with a small step is not a neutral reference for Lawson (MIS's own acoustic damping); RK4 on N is.

10 Open questions and next steps

- a1 terrain FV: identify the terrain term in S (Section 8).
- MIS3C: second order with $L \neq 0$; compare with Knoth & Wensch (2014).
- Cut cells: warm layer in the lowest cells downstream of the hill.
- From the MISDG plan: a high-order (Yoshida) composition fast integrator, which Lawson needs (Section 9); optionally ETDRK4 (talk in `docs/talks/misdg`).
- A linear IMEX fast solver as in the MATLAB reference (one sparse LU of $I - \gamma\Delta\tau L_I$), wider choices of the implicit part (adaptive masks), retained HDG traces in 2D.

References

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